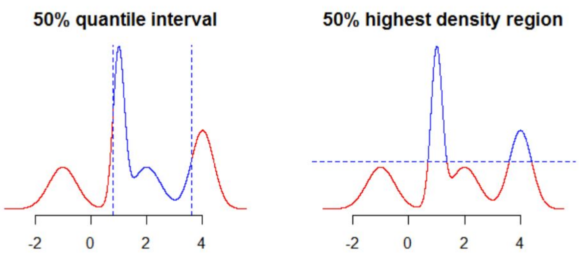
Lab 4

PSTAT 115

Highest posterior density region (HPD)



Posterior predictive distribution

- An important feature of Bayesian inference is the existence of a predictive distribution for new observations.
  - Let $\tilde{y}$ be a new (unseen) observation, and $y_1, ...y_n$ the observed data.
  - The Posterior predictive distribution is $p(\tilde{y} \mid y_1, ...y_n)$
- The predictive distribution does not depend on unknown parameters
- The predictive distribution only depends on observed data

The posterior predictive distribution allows us to find the probability distribution for new data given observations of old data.

$$
\begin{aligned}
p(\tilde{y} \mid y_1, ...y_n) &= \int p(\tilde{y}, \theta \mid y_1, ...y_n) d\theta \\
&= \int p(\tilde{y} \mid \theta) p(\theta \mid y_1, ...y_n) d\theta
\end{aligned}
$$

- The prior predictive distribution describes our uncertainty about a new observation before seeing data
- It incorporates uncertainty due to the sampling in a model $p(\tilde{y} \mid \theta)$ *and* our prior uncertainty about the data generating parameter, $p(\theta)$

Example

- $\lambda \sim \text{Gamma}(\alpha, \beta)$
- $\tilde{Y} \sim \text{Pois}(\lambda)$

$$
\begin{aligned}
p(\tilde{y}) &= \int p(\tilde{y} \mid \lambda) p(\lambda) d\lambda \\
&= \int (\frac{\lambda^{\tilde{y}}}{y!} e^{-\lambda})(\frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{(\alpha-1)} e^{-\beta\lambda}) d\lambda \\
&= \frac{\beta^\alpha}{\Gamma(\alpha) y!} \int (\lambda^{(\alpha+y-1)} e^{-(\beta+1)\lambda}) d\lambda
\end{aligned}
$$

$\int (\lambda^{(\alpha+y-1)} e^{-(\beta+1)\lambda}) d\lambda$ looks like an unormalized $\text{Gamma}(\alpha + y, \beta + 1)$

Sampling

Why We Need Different Sampling Strategies

- In Monte Carlo methods, we sometimes need to sample from a distribution. The pdf of this distribution can be of any form as long as it is a legal pdf.
- For some distributions, such as normal, beta, gamma and so on, we can easily do the sampling using build in functions in R.
- If not, we need to be more clever about how we generate samples. There are two common approaches for sampling from a *univariate* distribution:
  - Inversion Sampling
  - Rejection sampling

Probability Integral Transform

- Suppose that a random variable, Y has a continuous distribution for with CDF is F_Y.
- Then the random variable $U = F_Y(Y)$ has a uniform distribution
  - This is known as the "probability integral transform PIT"
- By taking the inverse of F_Y we have $F^{-1}(U) = Y$

Inversion Sampling

The inverse transform sampling method works as follows:

1. Generate a random number u from Unif$[0, 1]$
2. Find the inverse of the desired CDF, e.g. $F_Y^{-1}(u)$.
3. Compute $y = F_Y^{-1}(u)$. y is now a sample from the desired distribution.

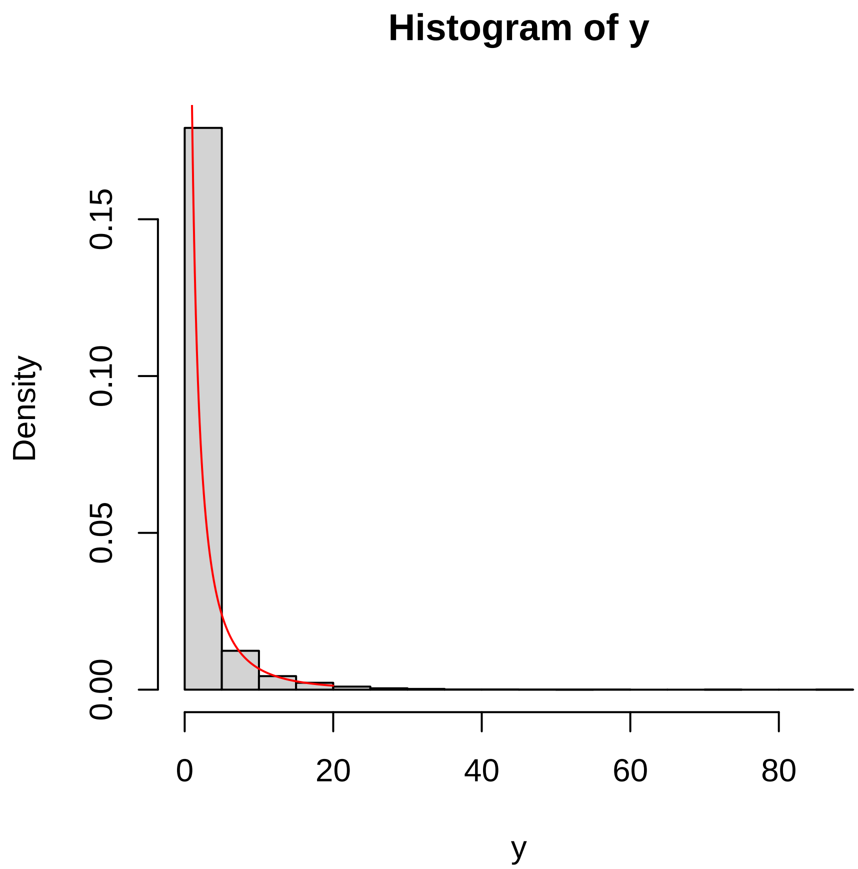
As an example, suppose we have our CDF function given by $F_Y(y) = 1 - e^{-\sqrt{y}}$. In order to perform an inversion we want to solve for $F(F^{-1}(u)) = u$.

$\Rightarrow 1 - e^{-\sqrt{F^{-1}(u)}} = u$

$\Rightarrow F^{-1}(u) = (log(1 - u))^2$

And with this equation, we can generate u from uniform distribution from $[0, 1]$ to get our y. We can plot a histogram on our generated y and then add the true density of y to see if everything is good.

```
set.seed(6666)
n <- 10^4
u <- runif(n)
y <- (log(1-u))^2
hist(y,freq=F)
d=seq(0.01,20,0.01)
lines(d,exp(-sqrt(d))/(2*sqrt(d)),type='l',col='red')
```


Histogram of y
Density
0.00
0.05
0.10
0.15
0
20
40
60
80
y